

5.6 CAD On Linux

You've probably been hearing of CAD for a long time now—a civil or mechanical engineer friend of yours keeps mentioning it. Well, “CAD” is short for Computer Aided Design. In simple words, it is the process or method of designing a physical entity or object, using simulation, on a computer. CAD has revolutionised the design process for buildings, automobiles, machinery, etc. A civil engineer can view the effect and appeal of a building with different structures, a mechanical engineer can verify if the various parts of a machinery match, and so on. CAD is important as it helps you understand complex structures by displaying them which you otherwise had to physically see or just visualise.

There are several CAD software out there, and we will mention some of them which are available for Linux. CAD software are generally complex to handle, and need some getting used to. While not all CAD software is available for free for Linux, you can still learn and get familiar with CAD with the free applications that are available. The basic operations will be more or less similar across the software.

One of the first software you can try out to get a hang of CAD is **BlenderCAD**, which is actually an extension to the Blender vector graphics application that we discussed earlier. You can get the source code from <http://projects.blender.org/projects/blendercad/> and compile it.

A simple program we can suggest is FreeCAD. The maker of FreeCAD, Askoh puts it as follows:

“FreeCAD allows users to create and manipulate assemblies of parts. The parts are simple 3D solids, which can be connected by joints, constraints, contacts, motors, actuators, springs, dampers, forces, torques or gravity. The parts and connections define the structure, mechanism or machine of interest. Both open and closed 3D loops are permitted. ‘freeCAD’ performs full Multibody Dynamics analysis on the assembly to predict the motion according to Newton’s Laws. Animation using the simulated data produces realistic dynamic behavior of the system.